

Review Article

PHYTOTHERAPY AND ITS ROLE IN THE TREATMENT OF DEPRESSION

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ABSTRACT:

For centuries, man has used traditional remedies for the treatment of various ailments. Despite all the marvelous advancements in the field of medical science, traditional medicine has always been in practice. More than 50% of the developing world's population still depends on complementary and alternative systems of medicine (CAM), not only due to their efficacy but also because many of the modern drugs are beyond the financial reach of general population. Evidence-based CAM therapies have shown remarkable success in healing acute as well as chronic diseases. These therapies are first choice for problems such as infertility, epilepsy, psychosomatic disorders, depression and many other ailments. Phytotherapy is one of the therapeutic modalities in CAM that is gradually becoming very popular. Herbal remedies and extracts are traditionally being used and available worldwide for the treatment of many diseases including depression. Many of these plants have been scientifically validated by experiments that justify their traditional use. The standardized extract of *Hypericum perforatum* (St. John's wort) is the most promising herbal remedy, used as a therapeutic agent for depression in many countries and it seems to offer significant advantages over conventional antidepressants. Herbal practitioners have claimed for centuries that extracts from plants are effectively used for the alleviation of different types of diseases. Here we have reviewed the scientific status of some popular herbs that are being used in different parts of the world for the treatment of depression.

PHYTOTHERAPY

With the beginning of human era, man's primary concern is health. Man is using different plants and herbs growing in his environment to treat various ailments since many centuries. From the use of plants and herbs growing in the environment, man was eventually able to create a system of therapeutics that dates back to 4000 years (SL, 1991).

Today phytotherapy, the use of herbs and herbal extracts for the treatment of diseases, is an established therapeutic modality although much of the science behind it is still in its infancy. Nevertheless, modern medicine also

benefited from the plants of therapeutic importance and many drugs are derived from natural sources (Table).

Phytochemical examination of plants available to earlier civilizations has often shown that many of these contain active principles, which are responsible for therapeutic success. Some compounds of plant origin may have limited application in clinical situations but are important tool for pharmacological research. Some examples are atropine from *Atropa belladonna*, an anti muscarinic agent, cocaine from *Erythroxylum coca* that inhibits catecholamine reuptake, reserpine from *Rawolfa serpentina*, depletes biogenic amines from nerve endings (Gilman,

2001). In the developing and third world countries, the uses of herbal remedies are very popular among the general population not only because they are effective but also because many modern medicines are beyond the financial reach of most of them. The situation is not different in the developed world also. Analysis of the twenty five best-selling pharmaceuticals in the world in 1993 and in the Dutch community in 1996 has shown that almost half of the total sales were from the drugs with a direct or indirect natural origin (De Smet, 1997). Out of the 520 new drugs approved in the period between 1983 to 1994 by either Food and Drug Authority (FDA) or comparable entities in other countries, there were 30 drugs obtained directly from natural origin, 173 were either semi synthetic from a natural product or synthetics originally modeled on a natural product (De Smet, 1997). There has been revival of interest in the natural products for the last three decades and the modern medicine is now beginning to accept the plants in crude form.

Phytotherapy in depression

Depression is a common mental health problem, which is associated with a high risk of mortality from comorbid medical illnesses and from suicide (Baldwin and Jolley, 1986).

It is characterized by disturbance of mood, insomnia or hypersomnia, fatigue or loss of energy, poor concentration, loss of sense of control and a subjective experience of great distress. It affects the thinking and functioning processes of an individual, greatly diminishes his or her social role and productivity (Licinio and Wong, 1999). The use of natural medicines including plants to control diseases is a centuries old practice, based on the experience of many generations of Traditional physicians and Herbal practitioners. Greeko-Arabic (Tibb-e-Unani), Ayurvedic (Indian) and other herbal medical practitioners have claimed for centuries that extract of plants can be effectively used for the alleviation of different types of diseases. For the last many years research on such plants took an upward surge throughout the world, to scientifically validate the traditional herbal remedies.

Natural products have played significant role in the management of neuropsychiatric disorders. There are many plants, which are used traditionally for the treatment of depression and their antidepressant activities have been proven by experiments. For example *Curcuma longa*, *Rhazya stricta* decoction, *Banxia houpu* decoction, *Bacopa monniera*, *Areca catechu*, *Apocynum venetum*,

Table
Examples of plant based drugs (Gilman, 2001)

PLANT	DRUG
<i>Atropa belladonna</i>	ATROPINE
<i>Chondrodendron tomentosum</i>	TUBOCURARINE
<i>Cinchona succirubra</i>	QUINIDINE, QUININE
<i>Digitalis purpurea</i> , <i>Digitalis lanata</i>	DIGOXIN, DIGITOXIN
<i>Ephedra sinica</i>	EPHEDRINE
<i>Papaver somniferum</i>	MORPHINE, CODEINE
<i>Physostigma venenosum</i>	PHYSOSTIGMINE
<i>Pilocarpus microphyllus</i>	PILOCARPINE
<i>Catharanthus roseus</i>	VINCRISTINE, VINBLISTINE
<i>Taxus brevifolia</i>	TAXOL
<i>Artemisia annua</i>	ARTEMISININE, ARTEMETHER
<i>Colchicum autumnale</i>	COLCHICINE
<i>Hyoscyamus niger</i>	HYOSINE
<i>Rawolfia serpentina</i>	RESERPINE

Morinda officinalis, *Hypericum perforatum*, *Psoralea corylifolia*, *Glycyrrhiza glabra*, *Trichilia catigua*, *Piper longum*, *Paullinia cupana*, *Epimedium brevicornum*, *Catuama*, *Hypericum canariense*, *Hypericum glandulosum*, *Chaihu-Shugan-San*, *Myristica fragrans* etc. We will cite above examples with a brief description of their pharmacological studies.

Curcuma longa (turmeric) is a well-known indigenous herbal medicine. The aqueous extracts, when administered orally to the mice from 140 to 560 mg/kg for 14 days, elicited immobility reduction in the tail suspension test (TST) and the forced swimming test (FST) in a dose dependant manner. The effects of the extracts at the dose of 560 mg/kg were more potent than that of reference antidepressant fluoxetine. The activity of *Curcuma longa* as antidepressant may be mediated in part through monoamine oxidase inhibition in mouse brain (Yu *et al.*, 2002).

Rhazya stricta decne leaves, which have both antidepressant and sedative properties in animal models, are widely used in folk medicine in the Arabian Peninsula. Rats were pretreated with various doses (0.025-6.4 mg/kg) of the lyophilized extract of the plant leaves, or with desipramine (10, 20 and 40 mg/kg) and were subjected to the forced swimming test. The result indicated that the plant extract caused reduction in immobility time at the dose of 0.8, 1.6 and 6.4 mg/kg in a dose dependent manner. Under the same experimental conditions desipramine (20 and 40 mg/kg) produced dose dependent decrease in immobility time. It is concluded that *Rhazya stricta extract* may possess an antidepressant like effect (Ali *et al.*, 1998).

Banxia houpu decoction, a traditional chinese medicinal formula which was prescribed 1000 years ago by Zhong jing Zhang, is being used in China to treat depression effectively. The scientific evaluation of this medicinal formula for antidepressant properties concluded that its

efficacy was similar to that of prozac, which is selective serotonin reuptake inhibitor (Luo *et al.*, 2000). The oral administration of polysaccharides from *Banxia-houpu decoction* exhibited reduction in the immobility time of rodents in the tail suspension and in the forced swimming tests in a dose dependent manner. The polysaccharides from *Banxia-houpu decoction* were active in animal models of depression with comparable effects to known antidepressants. The oral administration of polysaccharides at a low dose for 4 weeks resulted in a significant increase in a monoamine neurotransmitter 5-hydroxy tryptamine and dopamine levels in the whole mouse brain, but produced no significant increase in 5-hydroxyindolacetic acid (5-HIAA) and norepinephrine concentrations. The effect of polysaccharides on the brain neurotransmitter levels appeared to be quite different from the effect of fluoxetine (Guo *et al.*, 2004).

Bacopa monniera, is a commonly used Aurvedic drug for mental disorders. The extract of the plant when given in the dose of 20 and 40 mg/kg, orally once daily for 5 days was found to have significant antidepressant activity in forced swimming test and learned helplessness models of depression and was comparable to that of reference drug imipramine at the dose of 10-20 mg/kg (Sairam *et al.*, 2002).

Areca catechu, commonly known as Chalia in the Indo-Pakistan subcontinent, has been shown to possess CNS stimulant properties (Nadkarni, 1976). The aqueous ethanolic extract (4-80 mg/kg) caused a significant reduction in immobility time of rodents without effecting spontaneous motor activity (Dar and Khatoon, 1997). The aqueous ethanolic extract and the hexane and aqueous fractions inhibit monoamine oxidase in rat brain homogenates (Dar *et al.*, 1997). The dichloromethane fraction from *Areca catechu* was found to inhibit monoamine oxidase type A isolated from the rat brain with an IC₅₀ of 665±65.1 µg/mL. Studies with pharmacological models of depression, i.e., forced

swimming and tail suspension tests, indicated that it caused significant reduction in the immobility time in rodents similar to that of moclobemide (a selective inhibitor of MAO-A) without causing a significant change in motor performance (Dar and Khatoon, 2000).

Apocynum venetum is a wild shrub widely distributed in mid and north western China. An extract of the leaves of *Apocynum venetum* markedly shortened the immobility time of rats in a forced swimming test in a dose range of 30-125 mg/kg, indicating the possible antidepressant activity that was comparable to that of the tricyclic antidepressant imipramine (20 mg/kg) (Butterweck *et al.*, 2001).

The plant extract of *Morinda officinalis*, a Chinese traditional medicinal plant at a dose of 25-50 mg/kg, is similar to clinically effective antidepressant drug desipramine (5-10 mg/kg). In the forced swimming test in mice, the plant extract (50 mg/kg), like the effect of desipramine (20 mg/kg), elicited a significant reduction in the duration of immobility time. A tendency to this phenomenon could be seen at the dose of 100 mg/kg. Meanwhile, the plant extract, in the effective doses for the forced swimming test, had no effect on spontaneous motor activity in mice. These findings provide further support for the conclusion that *Morinda officinalis* extract possesses the antidepressant effect (Zhang *et al.*, 2002).

The extracts of *Hypericum perforatum* (St. John's wort) are used in the treatment of depression in many countries and represent an accepted alternative to available pharmacological antidepressants. St. John's wort extracts are therefore used as a therapeutic agent for the treatment of depression and it extends well beyond the traditional field of herbal medicine (Muller, 2003) and now allopathic medical practitioners are also prescribing St. John's wort for the same. This wort has been subjected to extensive scientific studies in the last decade and there are still many open questions about its pharmacology and mechanism of action. Analytical studies reported that St. John's wort is a non selective

inhibitor of monoamine oxidase-A and -B activity and it also inhibits the synaptosomal uptake of serotonin, dopamine and norepinephrine with approximately equal affinity (Butterweck, 2003). Biochemical studies suggest that the phloroglucinol derivative hyperforin is the main active ingredient of St. John's wort (Di Carlo *et al.*, 2001). No other antidepressant shows the similar sort of broad inhibitory profile as compared to this herb (Muller, 2003). St. John's wort has been shown to alleviate symptoms of mild to moderate depression, and seems to offer significant advantages over conventional antidepressants (Di Carlo *et al.*, 2001). This is because of very low incidence of adverse effects, almost similar to that of the placebo and consistently lower than that of the conventional antidepressant drugs. The most common adverse effects of St. John's wort are allergic skin reactions and very rare adverse effects included are gastrointestinal symptoms, confusion, tiredness, sedation, dry mouth, restlessness, headache, apathy, tremors, and photophobia (Ernst *et al.*, 1998). In view of its efficacy and safety records, St. John's wort should be considered for the first-line treatment of mild to moderate depression (Di Carlo *et al.*, 2001).

Psoralea corylifolia (Babachi) is used as folk medicine in many parts of India and China. It was found to reduce the duration of immobility time in the forced swimming test without accompanying changes in ambulatory, rearing and grooming behaviors in the open-field test. At the doses from 7.5 to 100 mg/kg, *Psoralea corylifolia* significantly decreased the immobility time of rodents. The efficacy of the higher doses exceeded that of amitriptyline at 10 and 20 mg/kg and fluoxetine at 13 mg/kg. In animal brain and liver, it significantly inhibited monoamine oxidase A and B activities (Chen *et al.*, 2006).

The aqueous extract of *Glycyrrhiza glabra*, popularly known as liquorice, (75, 150, and 300 mg/kg) was administered orally for 7 successive days in separate groups of Swiss young male albino mice. The dose of

150 mg/kg of the extract significantly reduced the immobility time of mice in both FST and TST, without any significant effect on locomotor activity of mice. The efficacy of extract was found to be comparable to that of imipramine (15 mg/kg i.p.) and fluoxetine (20 mg/kg i.p.). Further experimental work revealed that antidepressant-like effect of liquorice extract seems to be mediated by increase of brain norepinephrine and dopamine levels. Monoamine oxidase inhibiting effect of liquorice may be contributing favorably to the antidepressant like activity. Thus, it is concluded that liquorice extract may possess an antidepressant-like effect (Dhingra and Sharma, 2006).

Trichilia catigua is a Brazilian medicinal plant that is traditionally used for the treatment of depression. Acute oral treatment with the extract of *T. catigua* produced antidepressant like effects in the forced swimming model in both mice and rats. In vitro, the extract of this plant inhibited the uptake and increased the release of serotonin, and especially of dopamine, from rat brain synaptosomal preparations in a dose dependant manner. It was concluded that the active principle(s) present in the hydroalcoholic extract of *T. catigua* showed dopamine-mediated antidepressant-like effect in mice and rats (Campos *et al.*, 2005).

Piper longum is a medicinal plant. A bioassay guided isolation of the ethanol extract from the fruits of this plant yielded a known piperidine alkaloid, piperine that is a monoamine oxidase (MAO) inhibitor. Piperine significantly reduced the duration of immobility time of mice in the tail suspension test, similar to that of the reference antidepressant fluoxetine, without accompanying changes in ambulation when assessed in an open-field. These results suggest that piperine possesses potent antidepressant like properties that are mediated in part through the inhibition of MAO activity, and therefore represent a promising pharmacotherapeutic drug as an antidepressant agent (Lee *et al.*, 2005).

Guarana, a herbal extract from the seeds of *Paullinia cupana* significantly reduced the duration of immobility in the forced swimming test at the dose of 25 and 50 mg/kg, p.o. At these doses, it did not affect ambulation in the open field test, suggesting an antidepressant-like effect in mice (Campos (b) *et al.*, 2005).

Icariin that was isolated from *Epimedium brevicornum* was found to be significantly shortened the immobility time of rats in the forced swimming test (FST) after orally administration for 21 consecutive days. Icariin also produced a marked reduction in immobility time of mice in the tail suspension test (TST) when administered for at least 7 consecutive days. Moreover, it was observed that the stress of FST exposure induced increases in brain monoamine oxidase (MAO) A and B activities, serum corticotropin-releasing factor (CRF) levels, as well as decreases in brain monoamine neurotransmitter levels. These results suggested that icariin possessed potent antidepressant like properties that were mediated via neurochemical and neuroendocrine systems (Pan *et al.*, 2005).

Catuama is a herbal product currently used as a tonic, especially for the management of mental disorders. Acute and chronic oral treatments with *Catuama* resulted in a significant reduction of the immobility time of rodents in FST and TST both. Conversely, treatment with the same doses of *Catuama* did not significantly interfere with motor activity that was assessed by the open-field test. The antidepressant-like effects were comparable to those observed for classical antidepressant drugs. When assessed in vitro, *Catuama* inhibited the synaptosomal uptake of norepinephrine, serotonin and dopamine in rat brain in a concentration-dependent manner. Likewise, in vitro incubation of this plant also resulted in a marked increase of the release of serotonin and dopamine in rat brain crude preparation of synaptosomal membranes. Finally, *Catuama* was found to be effective in interfering with the synaptosomal uptake of serotonin and dopamine following long-term

oral treatment of rats. These findings scientifically validates the traditional use of this herbal product as an antidepressant (Campos *et al.*, 2004).

The butanol and chloroform fraction of *Hypericum canariense* and *Hypericum glandulosum* significantly reduced the duration of immobility time of mice in the forced swimming test, producing no significant change on spontaneous motor activity when assessed in a photocell activity meter. In this regard, the efficacy of the chloroform extract from *Hypericum glandulosum* (500 mg/kg p.o.) in the forced swimming test was comparable to that of the tricyclic antidepressant imipramine (50 mg/kg p.o.). These findings revealed that the butanol and chloroform fractions from *Hypericum canariense* and *Hypericum glandulosum* possess antidepressant-like effects in mice, providing further support for the traditional use of these plants for the treatment of depression (Sanchez-Mateo *et al.*, 2005).

Chaihu-Shugan-San (CSS), an oriental traditional medicinal plant which is used as a remedy for depression. The herbal extracts were administered orally for 21 consecutive days in mice and was compared with imipramine. This study concluded that CSS and its constituents exert antidepressant-like effects comparable to those of imipramine in experimental animal models (Kim *et al.*, 2005).

Myristica fragrans extract (5, 10, and 20 mg/kg) was administered orally for 3 successive days to different groups of Swiss male young albino mice. *M. fragrans* extract significantly decreased immobility periods of mice in both the FST and the TST. The 10 mg/kg dose was found to be most potent, as indicated by the greatest decrease in the immobility period compared with the control animals. Furthermore, this dose of the extract was found to have comparable potency to imipramine (15 mg/kg i.p.) and fluoxetine (20 mg/kg i.p.). The extract did not have a significant effect on locomotor activity of

mice. Further experiments revealed that antidepressant-like effect of the extract seems to be mediated by interaction with the adrenergic, dopaminergic, and serotonergic systems (Dhingra (b) and Sharma, 2006).

In the light of above cited studies, it may be safely said that phytotherapy does play an important role in the treatment of depression. We suggest, comparative studies are being required at local level where both herbal and current antidepressants must be used in patients to observe significant results of these therapies. It is possible that further studies may lead to an emergence of an excellent pharmacotherapeutic agent of herbal origin for this pandemic mental disorder.

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